Heart Transplant

ORG: S-535 (ISC)
Link to Codes

MCG Health
Inpatient & Surgical Care
28th Edition

Care Planning - Inpatient Admission and Alternatives

Clinical Indications for Procedure

- Procedure is indicated for **ALL** of the following(1)(2)(3)(4)(5)(6):
  - Transplant need, as indicated by **1 or more** of the following:
    - Cardiogenic shock or severe heart failure (eg, New York Heart Association class IV) requiring continuous intravenous inotropic support or mechanical cardiac support (eg, intra-aortic balloon pump, extracorporeal membrane oxygenation, ventricular assist device)(12)(13)(14)(15)
    - Severe chronic heart failure, as indicated by **ALL** of the following(15)(16)(17):
      - New York Heart Association class III or IV despite maximally tolerated guideline directed medical therapy (GDMT)
      - Peak metabolic oxygen consumption[A] on cardiopulmonary exercise test less than 14 mL/kg/min or less than 12 mL/kg/min if patient on beta-blocker
      - Peak metabolic oxygen consumption[A] on cardiopulmonary exercise test less than 50% predicted based on patient age, sex, and weight
    - Severe cardiac ischemia (eg, angina) despite maximal feasible therapy (eg, revascularization, medication)
    - Recurrent symptomatic or life-threatening arrhythmia unresponsive to medical therapy and interventional procedures (eg, catheter ablation, intracardiac defibrillator)(11)(17)
    - Severe congenital heart disease, as indicated by **1 or more** of the following(18)(19)(20)(21):
      - Single ventricle physiology
      - Failed Fontan circulation[B]
      - Eisenmenger syndrome
      - Reactive pulmonary hypertension with risk for progression to a level of fixed pulmonary vascular resistance that may preclude future transplant
      - Ventricular failure due to congenital heart disease that is not amenable to other surgical alternatives
      - Severe oxygen desaturations not amenable to other surgical correction
      - Severe heart failure refractory to medical therapy not amenable to other surgical, interventional, or electrophysiologic intervention
    - Low-grade myocardial tumor and **ALL** of the following:

            - No evidence of metastatic disease
            - Tumor is otherwise unresectable.
        - Unresectable ventricular diverticula
        - Retransplant for graft dysfunction from severe allograft vasculopathy
    - No contraindications to heart transplant present

Alternatives to Procedure

- Alternatives include(1)(2)(3)(4)(5)(16)(20):
    - Maximum medical management of heart failure
    - For fulminant myocarditis: corticosteroids and immunosuppressive therapy, discontinuation of causative agent (eg, drug hypersensitivity reaction, immune checkpoint inhibitor cardiotoxicity) if appropriate, or possible mechanical assist device(15)(23)
    - Coronary revascularization. See Coronary Artery Bypass Graft (CABG) ISC or Percutaneous Coronary Intervention ISC guideline.
    - Combined cardiac resynchronization therapy and implantable cardioverter-defibrillator.(16) See Electrophysiologic Study and Implantable Cardioverter-Defibrillator (ICD) Insertion ISC guideline.
    - Ventricular assist device (mechanical circulatory support) for(15)(24)(25):
        - Recovery support in postcardiotomy, postinfarction, or post-transplant "stunned" myocardium, or for myocarditis
        - Bridging support in transplant or retransplant recipients in life-threatening heart failure, arrhythmias, or end organ failure
        - Destination therapy for patient who is not candidate for heart transplant
    - Nontransplant surgical alternatives for heart failure, including:
        - Coronary artery revascularization
        - Intra-aortic balloon pump(12)(25)
        - Left ventricular resection
        - Cardiac valve replacement(26)
        - Left ventricular reconstruction
        - Creation of Fontan circulation or other procedures for complex congenital heart disease
        - Total artificial heart(15)(27)
    - Cardiac rehabilitation(28)
    - Palliative care(16)(28)(29)

Operative Status Criteria

Goal Length of Stay: 12 days postoperative

- Inpatient

Preoperative Care Planning

- Preoperative care planning needs may include(1)(2)(3)(4)(5)(16)(19)(20)(30):
    - Preoperative evaluation, including:
        - Thorough evaluation by multidisciplinary heart transplant team for:
            - Severity of heart disease
            - Alternatives to heart transplant
            - Contraindications to heart transplant
            - Psychosocial evaluation (eg, mental health, substance abuse history, treatment adherence, social support, social history, cognitive status, and knowledge of treatment and options)(31)(32)(33)
        - Protocol as specified by transplant center
        - Routine preoperative evaluation. See Preoperative Education, Assessment, and Planning Tool SR.
    - Diagnostic test scheduling, including(34):
        - Cardiovascular testing including right heart catheterization (with vasodilator testing if pulmonary hypertension present)
        - Pulmonary testing
        - Cardiopulmonary exercise test (to measure peak metabolic oxygen consumption and rule out pulmonary cause of exercise limitation)
        - Laboratory testing
        - Nutrition evaluation
        - Dental evaluation
        - Immunology screening, including blood and tissue typing
        - Detailed infectious disease screening, including viral antibody titers(35)(36)
        - Hemoglobin A1c(37)
    - Preoperative treatment, procedures, and stabilization, including:
        - Optimizing cardiac medication regimen
        - Intravenous inotropic agents(38)
        - Pulmonary vasodilator agents (eg, inhaled nitric oxide, prostacyclin)

  - Ventricular assist device (left, right, or biventricular mechanical circulatory support)(39)
  - Intra-aortic balloon pump(25)
  - Extracorporeal membrane oxygenation (ECMO)(40)
  - Total artificial heart
  - Pretransplant desensitization to HLA antibodies(41)
  - Smoking cessation counseling(37)
  - Alcohol cessation counseling(37)
- Preoperative discharge planning as appropriate. See Discharge Planning in this guideline.

Hospitalization

Optimal Recovery Course

Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
1	• OR to ICU • Social Determinants of Health Assessment • Discharge planning	• **Clinical Indications met[C]** • Intubated	• Bed rest • Turn side to side every 2 hours	• IV fluids • IV medications • NPO	• Central lines • Pacing wires • Continuous ECG monitoring • Oximetry • CBC, chemistries • CXR postoperatively • Possible extubation postoperatively • Oxygen • Chest tube	• Perioperative antibiotics • Immunosuppressive medications • Possible inhaled pulmonary vasodilators • Possible parenteral vasoactive medication
2	• ICU • Social Determinants of Health Assessment	• **Hypotension absent**	• Out of bed to chair with assistance if tolerated • Turn side to side every 2 hours	• Diet as tolerated • IV fluids • IV medications	• ECG monitoring • Temporary pacing if indicated • Oxygen	• Perioperative antibiotics • Immunosuppressive medications • Possible inhaled pulmonary vasodilators • Possible parenteral vasoactive medication
3	• ICU • Social Determinants of Health Assessment	• **Hemodynamic stability** • **Mechanical ventilation absent**	• **Dangle legs** • Out of bed to chair with assistance	• Diet as tolerated • IV fluids • IV medications	• ECG monitoring • Temporary pacing if indicated • Oxygen	• **Wean parenteral vasopressor medication** • Possible inhaled pulmonary vasodilators • Antibiotics absent • Immunosuppressive medications • Aspirin
4	• ICU or Intermediate care • Social Determinants of Health Assessment	• **Dangerous arrhythmia absent**	• Up to chair twice a day	• **Clear liquid diet** • IV fluids • IV medications	• ECG monitoring • Temporary pacing if indicated • Oxygen	• **Parenteral vasopressor medication absent** • Immunosuppressive medications • Aspirin

5	• Intermediate care • Social Determinants of Health Assessment	• **Pain absent or managed**	• **Limited ambulation**	• **Full liquid diet** • Oral medications • IV fluids	• **Chest tube absent** • ECG monitoring • Temporary pacing if indicated • Oxygen	• Immunosuppressive medications • Aspirin
6	• Intermediate care • Social Determinants of Health Assessment	• **SaO_2 greater than 90% or at baseline**	• Advance activity as tolerated	• Oral diet as tolerated • Oral medications • Oral or IV hydration	• **Oxygen absent** • Endomyocardial biopsy	• Immunosuppressive medications • Aspirin
7	• Intermediate care or floor • Social Determinants of Health Assessment	• **Pulmonary edema absent or mild**	• Advance activity as tolerated	• Oral diet as tolerated • Oral medications • Oral or IV hydration	• **Temporary pacing absent**	• Immunosuppressive medications • Aspirin
8	• Intermediate care or floor • Social Determinants of Health Assessment	• **Peripheral edema absent or mild**	• Advance activity as tolerated	• Oral diet as tolerated • Oral medications • Oral or IV hydration	• **Pacer wires absent**	• Immunosuppressive medications • Aspirin
9	• Floor • Social Determinants of Health Assessment	• **Stable fluid balance**	• **Ambulatory**	• Oral hydration • Oral medications • Oral diet	• **Central lines absent**	• Immunosuppressive medications • Aspirin
10	• Floor • Social Determinants of Health Assessment	• **Adequate diet tolerated**	• Ambulatory	• **Oral hydration** • **Oral medications** • **Oral diet**		• Immunosuppressive medications • Aspirin
11	• Floor • Social Determinants of Health Assessment	• **Biopsy not indicated or stable post biopsy** • **No evidence of organ rejection**	• Ambulatory	• Oral hydration • Oral medications • Oral diet		• Immunosuppressive medications • Aspirin
12	• Floor • Social Determinants of Health Assessment	• **Heart rate acceptable** • **No evidence of postoperative or surgical site infection**	• Ambulatory	• Oral hydration • Oral medications • Oral diet	• Sternotomy incision open to air	• Immunosuppressive medications • Aspirin

13	• Social Determinants of Health Assessment • Floor to discharge • Complete discharge planning	• **Adequate diet tolerated** • **No evidence of organ rejection** • **Pulmonary edema absent or mild** • **Immunosuppression regimen appropriate for next level of care** • **Cardiac function and fluid balance appropriate for next level of care** • **Dangerous arrhythmia absent** • **Hemodynamic stability** • **Pain absent or managed** • **Discharge plans and education understood**	• **Ambulatory or acceptable for next level of care**	• **Oral hydration[D]** • **Oral medications or regimen acceptable for next level of care** • **Oral diet or acceptable for next level of care**		• Immunosuppressive medications • Aspirin
(19)(30)(37)(42)(43)						

Recovery Milestones are indicated in **bold**.

Goal Length of Stay: 12 days postoperative

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Extended Stay

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for(43):
  - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
    - Expect brief stay extension.
  - Severe pretransplant condition (eg, ventricular assist device, intra-arterial balloon pump, extracorporeal membrane oxygenation (ECMO))(45)(46)(47)
    - Expect brief to moderate stay extension.
  - Complications of procedure(12)(30)(48)
    - Hemorrhage, infection, pulmonary hypertension, stroke, and multiple organ failure may occur and require medical and possibly surgical management.
    - Stay extension varies.
  - Primary allograft dysfunction(47)(49)(50)
    - Anticipate inotropic agents, possible ventricular assist device, extracorporeal membrane oxygenation, or other mechanical support.
    - Retransplant may be indicated for patients who do not recover cardiac function and have low-risk factors for recurrence.
    - Expect moderate to prolonged stay extension.
  - Acute rejection
    - Anticipate endomyocardial biopsy and serum assays for donor-specific antibodies.
    - Patients may require cytotoxic, antibody-directed, or immunosuppressive therapy, inotropic agents, and possible ventricular assist device, ECMO, or other mechanical support.
    - Expect moderate stay extension.
  - Post-transplant development of hemodynamic instability (vasoplegia), respiratory failure, or renal insufficiency(30)(51)
    - Expect brief to moderate stay extension.
  - Post-transplant arrhythmia
    - Anticipate medical and device (eg, pacemaker) treatment of arrhythmia.
    - Expect brief stay extension.
  - Active comorbidities

        - Patient with unstable comorbidities, such as obstructive or restrictive lung disease, pretransplant hepatic or renal insufficiency, pulmonary hypertension, diabetes mellitus, or cerebral or peripheral vascular disease, may require continued inpatient care.
        - Expect brief to moderate stay extension.
    - Severe obesity (BMI greater than 38)(52)
        - Patients with severe obesity have higher risk for postoperative stroke, pacemaker placement, and need for dialysis.
        - Expect brief stay extension.
    - Transplant of more than one organ (eg, heart-lung, heart-kidney, heart-liver)
        - Expect moderate to prolonged stay extension.

See Common Complications and Conditions ISC for further information.

Hospital Care Planning

- Hospital evaluation and care needs may include(19)(30)(37)(42):
    - Diagnostic test scheduling and completion, including:
        - Laboratory tests (eg, CBC, electrolytes, biomarkers, immunosuppressant levels)
        - Imaging (eg, serial chest x-ray, myocardial perfusion imaging)(53)
        - Echocardiogram(54)
    - Treatment and procedure scheduling and completion, including:
        - IV antibiotics
        - Inotropic support
        - Temporary pacemaker
        - Endomyocardial biopsy
        - Thoracentesis
        - Establishment of immunosuppression protocols(55)
        - Glycemic control(37)
    - Consultation, assessment, and other services scheduling and completion, including:
        - Medical and surgical specialty consultations as indicated
    - Monitoring patient's status for deterioration and comorbid conditions (see Inpatient Monitoring and Assessment Tool SR); key items include:
        - Hemodynamic stability measurements, which may include(29):
            - Blood pressure via arterial line
            - Pulmonary artery pressure
            - Central venous pressure
            - Cardiac output and cardiac index
            - Urine output
        - Need for vasoactive drugs
        - Need for pulmonary vasodilator agents (eg, nitric oxide, inhaled prostacyclin)
        - Respiratory function and oxygenation
        - Need for chest tube
        - Cardiovascular status, including presence of:
            - Arrhythmias
            - Electrolyte imbalances, especially potassium and magnesium
            - Signs and symptoms of cardiac tamponade, bleeding, hypertension, or hypovolemia
        - Wound and sternum tenderness, swelling, or serous drainage from incision
        - Pain management
        - Renal function
        - Gastrointestinal status
        - Signs and symptoms of infection, including:
            - Fever higher than 100.4 degrees F (38 degrees C)
            - Hypotension
            - Elevated cardiac output and cardiac index
            - Low systemic vascular resistance
            - Elevated WBC count
            - Purulent drainage from site of infection
            - Increasing oxygen requirements
            - Mental status change

Discharge

Discharge Planning

- Discharge planning includes[E]:

- Assessment of needs and planning for care, including(57)(58):
  - Develop and modify treatment plan (involving multiple providers) as needed.
  - Evaluate and address preadmission functioning as needed.
  - Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment SR for further information.
  - Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool SR for further information.(56)
  - Evaluate and address patient or caregiver preferences as indicated.
  - Identify skilled services needed at next level of care, with specific attention to:
    - Cardiovascular status assessment(59)
    - Medication management, adherence instruction, and side effects assessment(60)
    - Ongoing education required(61)
    - Psychosocial assessment, management, and referrals(59)
    - Wound or dressing management
- Early identification of anticipated discharge destination; options include(58)(62):
  - Home, considerations include:
    - Home safety assessment. See Home Safety Assessment SR for further information.
    - Patient safe to go home; examples include(63)(64)(65):
      - Medical status stable for patient's condition
      - Functional care can safely be provided with available resources.
      - Mental status stable for patient's condition
      - Medication availability confirmed and reconciliation complete
      - Patient/caregiver education completed with written discharge instructions provided
      - Community resources identified and referrals made, as needed
      - Home care arranged, if indicated
      - Necessary medical equipment delivery arranged or available in home, if indicated
      - Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated
    - Access to follow-up care
    - Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment SR for further information.
    - Caregiver need, ability, and availability
  - Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool SR for further information.
- Transitions of care plan complete, including(58)(62)(66):
  - Patient and caregiver education complete. See Heart Transplant: Patient Education for Clinicians SR for further information.
    - See Teach Back Tool SR for further information.
  - Medication reconciliation completion includes(60)(67):
    - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
    - Assess each medication for correlation to disease state or medical condition.
    - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
    - Provide reconciled medication list to all treating providers.
    - Confirm that patient or caregiver can acquire medication.
    - Educate patient and caregiver.
      - Provide complete medication list to patient and caregiver.
      - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
      - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(68)
    - Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
    - See Medication Reconciliation Tool SR for further information.
  - Plan communicated to patient, caregiver, and all members of care team, including(69):
    - Inpatient care and service providers
    - Primary care provider
    - All post-discharge care and service providers
  - Appointments planned or scheduled, which may include:
    - Primary care provider
    - Cardiologist(59)
    - Cardiothoracic surgeon(70)
    - Rehabilitation therapy services(71)(72)

        - Transplant center or team(73)
        - Other
    - Outpatient testing and procedure plans made, which may include:
        - Laboratory testing(70)
        - Other
    - Referrals made for assistance or support, which may include:
        - Community services(73)
        - Educational program (eg, chronic condition management, self-management)(74)
        - Financial, for follow-up care, medication, and transportation
        - Self-help or support groups
        - Tobacco use treatment(75)
        - Vocational rehabilitation(71)
        - Other
    - Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include:
        - Antiembolic or compression stockings
        - Blood pressure cuff
        - Weight scale
        - Wound care equipment and supplies(70)(76)
        - Other

Discharge Destination

- Post-hospital levels of admission may include:
    - Home.
    - Home healthcare. See Home Care Indications for Admission Section HC in Heart Transplant guideline in Home Care.

Evidence Summary

Criteria

The evidence for the clinical indications found in this guideline includes 15 published peer reviewed articles, 4 specialty society or other evidence-based guidelines, and 2 book sections.

Heart-lung transplant may be performed for patients with irreversible pulmonary hypertension, New York Heart Association class III or IV heart failure and end-stage lung disease, certain congenital heart diseases, or Eisenmenger syndrome not amenable to surgical correction or medical therapy.(7) **(EG 2)** Heart-liver transplant may be performed for patients with advanced heart failure and concomitant cirrhosis, Fontan circulation with severe Fontan-associated liver disease, selected patients with cardiac and hepatic amyloidosis, or other associated liver disease that may affect the cardiac allograft.(8)(9)(10) **(EG 2)** Heart-kidney transplant may be performed for patients with advanced heart failure and concomitant renal failure.(10) **(EG 2)**

For cardiogenic shock requiring inotropic or mechanical circulatory support, or severe symptomatic chronic heart failure despite maximally tolerated guideline directed medical therapy, a cardiovascular medicine textbook, an expert consensus statement, and a narrative review support heart transplant.(2)(3)(5) **(EG 2)**

For recurrent symptomatic or life-threatening arrhythmia unresponsive to medical therapy and interventional procedures (eg, catheter ablation, intracardiac defibrillator), an expert consensus statement and narrative reviews support heart transplant.(2)(5)(11) **(EG 2)**

For severe congenital heart disease, such as those with failed Fontan circulation or reactive pulmonary hypertension with risk for progression to a level of fixed pulmonary vascular resistance that may preclude future transplant, a specialty society guideline recommends heart transplant.(1) **(EG 2)**

For retransplant for graft dysfunction from severe allograft vasculopathy, a specialty society guideline, a cardiovascular medicine textbook, and a narrative review support heart transplant.(1)(2)(3) **(EG 2)**

Alternatives

Patients with advanced heart failure can show improvement via maximizing medical therapy and, when applicable, using cardiac resynchronization therapy, revascularization, ventricular resection, or valve repair or replacement.(15)(16) **(EG 2)** Placement of a left ventricular assist device (LVAD) as long-term (destination therapy) can be an alternative or bridge to transplant.(3)(4)(5)(16) **(EG 2)** Review of a national database included 25,621 patients (median age 58 years) with cardiogenic shock who received venoarterial extracorporeal membrane oxygenation (VA-ECMO) and found that the mortality rate was 56% and only 2.5% went on to receive a heart transplant while 6% received a left ventricular assist device and 36% recovered without use of a left ventricular assist device or heart transplant.(22) **(EG 2)**

Length of Stay

Analysis of a cohort of 255 adult patients who underwent heart transplant found a median length of stay of 9 days.(44) **(EG 2)** Analysis of national hospital discharge data shows 23% of hospitalized adult patients undergoing the principal procedure of heart transplant discharged in 12 days or less.(43) **(EG 3)**

Rationale

Use of this MCG care guideline helps the clinician identify, for a given procedure, which patient-specific factors and clinical conditions are appropriate for that procedure. The evidence-based clinical indication criteria assist the clinician in the decision to appropriately perform a procedure, evaluating whether the potential benefits of a procedure outweigh the potential risks. For Medicare enrollees, surgical MCG care guidelines also identify which procedures CMS has designated as inpatient only.

Use of these evidence-based clinical criteria to support decision making around the need for a given procedure is of benefit to the patient, as all procedures come with inherent risk that must be balanced by anticipated clinical benefit. Utilizing evidence-based clinical criteria enables a more accurate and patient-specific decision-making process. In addition, the use of evidence-based guidelines can help reduce unwarranted variation in care, such as divergent clinical thresholds to perform a procedure for clinically similar patients that vary across geographic regions, between facilities, and among individual clinicians.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 412.3(77); 42 CFR 419.22(78); 42 CFR 422.101(79)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 1 - Inpatient Hospital Services Covered Under Part A(80); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 6 - Hospital Services Covered Under Part B(81); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(82); CMS IOM Publication 100-08, Medicare Program Integrity Manual, Chapter 6, Section 6.5 - Medical Review of Inpatient Hospital Claims for Part A Payment(83)

Medicare Coverage Determinations: Medicare Coverage Database(84)

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Footnotes

[A] Peak metabolic oxygen consumption should be measured on patients receiving maximally tolerated guideline directed medical therapy (GDMT).(4) [A in Context Link 1, 2]

[B] Failed Fontan circulation may be an indication for combined heart-liver transplant.(9) [B in Context Link 1]

[C] See Clinical Indications for Procedure in this guideline. [C in Context Link 1]

[D] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [D in Context Link 1]

[E] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(56) [E in Context Link 1]

Definitions

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):
  - Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
  - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
  - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
  - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
  - Coma (ie, not arousable)

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Dangerous arrhythmia absent

- Dangerous arrhythmia absent, as indicated by absence of **ALL** of the following(1)(2)(3)(4)(5)(6)(7)(8)(9)(10)(11):
  - Resuscitated ventricular fibrillation or cardiac arrest
  - Ventricular escape rhythm
  - Sustained (30 seconds or more of ventricular rhythm at greater than 100 beats per minute) ventricular tachycardia
  - Nonsustained ventricular tachycardia with myocarditis or ischemia
  - Type II second-degree atrioventricular block
  - Third-degree atrioventricular block
  - New-onset left bundle branch block with suspected myocardial ischemia
  - Heart rhythms of concern due to association with Hypotension, Respiratory distress, or other significant symptoms (eg, syncope)

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Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:

- Hemodynamic abnormalities at baseline or acceptable for next level of care
- Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
    - Tachycardia absent
    - Hypotension absent
    - No evidence of inadequate perfusion (eg, no myocardial ischemia)
    - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

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Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4):
    - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
    - Low blood pressure, as indicated by **1 or more** of the following:
        - SBP less than 90 mm Hg in adult or child 10 years or older[A]
        - Mean arterial pressure[A][B] less than 70 mm Hg in adult or child 10 years or older
        - Decrease in baseline mean arterial pressure of 25% or more,[A][B] with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)
        - SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age[A]
        - SBP less than 70 mm Hg in infant 1 to 11 months of age[A]

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Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP. A calculator is available at https://www.mdcalc.com/mean-arterial-pressure-map.

Hypotension absent

- Hypotension absent[A], as indicated by **1 or more** of the following(1)(2)(3)(4):
    - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
    - Mean arterial pressure[B] greater than or equal to 70 mm Hg in adult or child 10 years or older
    - Mean arterial pressure[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
    - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
    - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

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Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP.

Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following(1):
  - Patient without baseline need for supplemental oxygen with oxygen saturation (SaO_2) of less than 90% or arterial blood gas partial pressure of oxygen (PO_2) of less than 60 mm Hg (8.0 kPa) on room air[A][B]
  - Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SaO_2 greater than 89% or PO_2 greater than 59 mm Hg (7.9 kPa)[A]
  - Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

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Footnotes

A. Pulse oximetry (SpO2) may underestimate arterial saturation (SaO2), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.(2) Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.(2) Specific target values may require adjustment when care is delivered at high altitude.(1)(3)
B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

No contraindications to heart transplant present

- No contraindications to heart transplant present,[A] as indicated by **ALL** of the following(1)(3)(4)(5)(6)(7)(8)(9)(10)(11):
  - No severe systemic illness limiting life expectancy to less than 2 years (eg, malignancy, AIDS, autoimmune disease, Duchenne muscular dystrophy)(11)(12)(13)
  - No congenital anomaly that would preclude successful cardiac transplant (eg, severe pulmonary artery hypoplasia, severe pulmonary vein stenosis or aplasia)(14)
  - No irreversible severe hepatic dysfunction if considered for only heart transplant (eg, INR greater than 1.5 off warfarin)
  - No severe renal insufficiency if considered for only heart transplant (eg, GFR less than 30 mL/min/1.73m^2 (0.5 mL/sec/1.73m^2))(10)(11)
  - No severe obstructive pulmonary disease (eg, FEV_1 less than 1 L/minute)[B]

- No severe fixed (ie, pharmacologically irreversible) pulmonary hypertension (may include pulmonary veno-occlusive disease)[B] (13)
- No psychiatric illness or severe cognitive disability (eg, dementia) that may interfere with postoperative medical adherence
- No severe neurologic disorder inhibiting postoperative recovery (eg, severe amyloid peripheral neuropathy, severe dysautonomia)(11)
- No active substance abuse[C][D](2)(15)
- No active infection (except device-related infection in ventricular-assist device recipients)
- No clinically severe symptomatic cerebrovascular disease
- No active peptic ulcer disease
- No other potentially reversible contraindications (eg, poorly controlled severe hypertension)

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Footnotes

A. Specialty guidelines and narrative reviews suggest that right heart catheterization be performed every 3 to 6 months to reassess pertinent findings.(1)(2)
B. This contraindication may not be applicable to patients undergoing a combined heart-lung transplant.
C. Recipient eligibility regarding substance abuse may vary between transplant programs and may be based on health insurer requirements as well as state legislation.(15)
D. A structured drug or alcohol rehabilitative program may be appropriate in patients with a recent history of alcohol or drug abuse.(2)

Orthostatic hypotension

- Orthostatic hypotension,[A][B] as indicated by **1 or more** of the following(1)(2)(3):
  - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
  - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

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Footnotes

A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.(1)(2)(3)
B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Respiratory distress

- Respiratory distress, as indicated by **ALL** of the following(1)(2):
  - Patient with **1 or more** of the following:
    - Dyspnea (difficulty breathing)
    - Tachypnea
    - Abnormal breathing pattern (eg, chest retractions)
    - Other evidence of difficulty breathing
  - Evidence of respiratory compromise, as indicated by **1 or more** of the following:
    - Hypoxemia
    - Altered mental status
    - Other evidence of respiratory compromise (eg, respiratory acidosis)

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Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health(1)(2)(3)(4):
  - Housing insecurity, as indicated by **1 or more** of the following:
    - Individual or caregiver's current living situation is **1 or more** of the following(5):
      - Does not have own housing (eg, staying in a hotel, shelter, or with others)
      - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
      - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
    - Current housing has **1 or more** of the following:
      - Electrical appliances (eg, stove, refrigerator) not working or unavailable
      - Insufficient heating or cooling
      - Insufficient ventilation
      - Lead paint or pipes
      - Mold
      - Pests (eg, bugs) or rodents
      - Smoke detectors not working or unavailable
  - Food insecurity, as indicated by **1 or more** of the following(6):
    - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
    - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.
  - Insufficient transportation, as indicated by **1 or more** of the following(7):
    - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.

    - In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.
  - Insufficient utilities, as indicated by **1 or more** of the following(8):
    - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
    - In the past year, electric, water, gas, or oil company threatened to shut off services.
  - Personal safety risk, as indicated by **2 or more** of the following(6):
    - Individual is sometimes or frequently physically hurt by another person (including family member).
    - Individual is sometimes or frequently insulted or talked down to by another person (including family member).
    - Individual is sometimes or frequently threatened with physical harm by another person (including family member).
    - Individual is sometimes or frequently screamed or cursed at by another person (including family member).
  - Insufficient dependent care, as indicated by **1 or more** of the following:
    - In the past year, individual or caregiver was unable to work due to lack of dependent care.
    - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
    - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
    - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
  - Depression risk, as indicated by **ALL** of the following:
    - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
    - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

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Tachycardia absent

- Tachycardia[A][B] absent, as indicated by **1 or more** of the following(1)(2):
  - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
  - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
  - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
  - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
  - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
  - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

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Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachypnea

- Tachypnea,[A][B] as indicated by respiratory rate of **1 or more** of the following(2)(3):
  - Greater than 18 breaths per minute in adult or child 13 years of age or older[A]
  - Greater than 22 breaths per minute in child 6 to 12 years of age[A]
  - Greater than 25 breaths per minute in child 3 to 5 years of age[A]
  - Greater than 30 breaths per minute in child 1 or 2 years of age[A]
  - Greater than 40 breaths per minute in infant 6 to 11 months of age[A]
  - Greater than 45 breaths per minute in infant 3 to 5 months of age[A]
  - Greater than 60 breaths per minute in infant 1 or 2 months of age[A]

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3. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. A mildly increased respiratory rate (eg, 20 to 22 in an adult), absent other indications of serious illness (eg, hypoxemia, metabolic acidosis, decreased tidal volume, pain), may be transitory and not clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachypnea. Whether a respiratory rate above the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath to breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.(1) This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.(1)

Codes

ICD-10 Diagnosis: C38.0, D15.1, I09.81, I42.0, Q24.8, T86.22, T86.290, T86.32

ICD-10 Procedure: 02YA0Z0, 02YA0Z1, 02YA0Z2

CPT®: 33945

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